Initialisation and reproducibility

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable initialization and replay record.

Trace starts,explain symmetry and preserve actual values plus full experimental settings.

Keep one habit: W1–W3/W5 → selected new start comparison → project replay ledger → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Regularisation and early stopping. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Learning rate experiments. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Updating parameters. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Return to h = ReLU(x), y = vh, **training** pairs (1, 2) and (2, 4), ordinary MSE and a frozen hidden unit. Only v is trained. Keep full-data updates, learning rate 0.1 and exactly two updates. Compare initial v values 0, 1 and 3.

  This experiment changes only the start.

- The gradient is 5(v - 2), so each step gives new v = 0.5v + 1. Signed distance from 2 halves on every update.

  Different starting distances therefore produce different losses after this finite budget, even though all three runs approach the same minimum if this exact procedure continues.

- The equal losses for starts 1 and 3 arise because they are equally distant from 2. They do not have equal final parameters.

  This simple convex example does not show the multiple solutions or inactive units that can occur in larger networks.

Regularisation and early stopping

> **My remembered idea, example and correction**

Learning rate experiments

> **My remembered idea, example and correction**

Updating parameters

> **My remembered idea, example and correction**

Lesson reference for workbook practice

Initial v	After one update	After two updates	Final training MSE
0	1	1.5	0.625
1	1.5	1.75	0.15625
3	2.5	2.25	0.15625

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main onlyvtrained, hReLU(x), pairs(1, 2)/(2, 4), MSE 2.5(v−2)², rate.1, two updates; start 0 path0/1/1.5 loss.625, start 1 path1/1.5/1.75 loss.15625, start 3 path3/2.5/2.25 loss.15625. Random(seed 17)fouruniformdrawsare separate, not usedin those scalar runs. Samecallsrepeatwithinsetup; anextrauniformpreviewshiftsassignment. Separate pictured(a−2)², rate.25, start 0→1→1.5/start 4→3→2.5, bothloss.25.

  Symmetryy=aReLU(ux)+bReLU(vx), x1, t1, params[u, v, a, b]all 1, gradientsall 2, rate.1→all.8, y1.28, loss.0784. Allzerogradients0 underdeclaredcornerconvention. Save actual initial values/order and complete protocol, not seed alone.

- For the scalar lesson model, start v = -2 and apply two updates at rate 0.1. Give all three v values and the final **training** MSE.

> **My reasoning and evidence**
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MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main onlyvtrained, hReLU(x), pairs(1, 2)/(2, 4), MSE 2.5(v−2)², rate.1, two updates; start 0 path0/1/1.5 loss.625, start 1 path1/1.5/1.75 loss.15625, start 3 path3/2.5/2.25 loss.15625. Random(seed 17)fouruniformdrawsare separate, not usedin those scalar runs. Samecallsrepeatwithinsetup; anextrauniformpreviewshiftsassignment. Separate pictured(a−2)², rate.25, start 0→1→1.5/start 4→3→2.5, bothloss.25.

  Symmetryy=aReLU(ux)+bReLU(vx), x1, t1, params[u, v, a, b]all 1, gradientsall 2, rate.1→all.8, y1.28, loss.0784. Allzerogradients0 underdeclaredcornerconvention. Save actual initial values/order and complete protocol, not seed alone.

- The lesson runs starting at 1 and 3 finish with the same **training** MSE. Give their final v values and explain the equality without calling the models identical.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main onlyvtrained, hReLU(x), pairs(1, 2)/(2, 4), MSE 2.5(v−2)², rate.1, two updates; start 0 path0/1/1.5 loss.625, start 1 path1/1.5/1.75 loss.15625, start 3 path3/2.5/2.25 loss.15625. Random(seed 17)fouruniformdrawsare separate, not usedin those scalar runs. Samecallsrepeatwithinsetup; anextrauniformpreviewshiftsassignment. Separate pictured(a−2)², rate.25, start 0→1→1.5/start 4→3→2.5, bothloss.25.

  Symmetryy=aReLU(ux)+bReLU(vx), x1, t1, params[u, v, a, b]all 1, gradientsall 2, rate.1→all.8, y1.28, loss.0784. Allzerogradients0 underdeclaredcornerconvention. Save actual initial values/order and complete protocol, not seed alone.

- Both runs use the same **seed** and generator. Run A assigns the first four draws to its weights. Run B draws a preview number first, then assigns the next four. Must the initial weight arrays match? What should be saved?

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main onlyvtrained, hReLU(x), pairs(1, 2)/(2, 4), MSE 2.5(v−2)², rate.1, two updates; start 0 path0/1/1.5 loss.625, start 1 path1/1.5/1.75 loss.15625, start 3 path3/2.5/2.25 loss.15625. Random(seed 17)fouruniformdrawsare separate, not usedin those scalar runs. Samecallsrepeatwithinsetup; anextrauniformpreviewshiftsassignment. Separate pictured(a−2)², rate.25, start 0→1→1.5/start 4→3→2.5, bothloss.25.

  Symmetryy=aReLU(ux)+bReLU(vx), x1, t1, params[u, v, a, b]all 1, gradientsall 2, rate.1→all.8, y1.28, loss.0784. Allzerogradients0 underdeclaredcornerconvention. Save actual initial values/order and complete protocol, not seed alone.

- A forward-pass program returns the same wrong answer every run with a fixed **seed**. Has repeatability verified the implementation? Name two stronger checks.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use J(a) = (a - 3) squared, gradient 2(a - 3), rate 0.25 and two updates. Restart from a = 1, then restart from a = 5. Calculate both traces and final losses. Explain what can be concluded about these two runs.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Transfer the starting-value experiment independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled **initialisation** and reproducibility **evidence** sheet to Image-pattern Network Investigation. Show how you can trace how starting values affect a fixed **training** budget.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Distinguish equality of loss and parameters for **Initialisation** and reproducibility. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Initialisation** and reproducibility. Proposed explanation: "The same **seed** specifies the entire experiment". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use J(a) = (a - 3) squared, gradient 2(a - 3), rate 0.25 and two updates. Restart from a = 1, then restart from a = 5. Calculate both traces and final losses. Explain what can be concluded about these two runs.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Reproduce two **training** traces by saving their distinct initial values.
- Loss $J(a)=(a-2)^2$; gradient 2(a-2); rate 0.25; two updates. Restart from a = 0 and separately a = 4.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- Does the matching final loss mean the parameter values are equal? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-5/17

Initialisation and reproducibility

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Start from 0

Updates: a = 1 then 1.5; final loss 0.25.

2. Start from 4

Updates: a = 3 then 2.5; final loss 0.25.

3. Record experiment

Save initial values, rate, objective and update count; the same final loss hides different parameter paths.

CHANGE ONE THING

Repeat the a = 0 run with identical arithmetic and settings.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project milestone

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Reproduce two **training** traces by saving their distinct initial values.
- Which labelled step matches this detail? Save initial values, rate, objective and update count; the same final loss hides different parameter paths.
- 1. Record experiment
- 2. Start from 0
- 3. Start from 4
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Record experiment Save initial values, rate, objective and update count; the same final loss hides different parameter paths.

Initialisation supplies the parameter values before learning begins. The same update rule can follow different paths from different starts.

Here the two paths end on opposite sides of two and have equal squared distance, so equal loss does not imply equal parameters. Save initial values and settings to reproduce this arithmetic, as well as the final result.

Changed-case question and optional hint

- Changed-case practice: Repeat the a = 0 run with identical arithmetic and settings. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. It again reaches 1 then 1.5 in this deterministic toy calculation.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: It again reaches 1 then 1.5 in this deterministic toy calculation. Identical initial state and update rules determine the same arithmetic trace.

Repair and retain the record

Review Regularisation and early stopping (ai-5/16). Goal: Separate a weight penalty from **prediction** error. Short practice: Why should the penalty value not be reported as validation error?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learning rate experiments (ai-5/14). Goal: Run a controlled comparison of learning rates. Short practice: What should a fair rate comparison hold fixed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Updating parameters (ai-5/11). Goal: Apply a learning rate to an ordered gradient. Short practice: Why must new loss be recomputed rather than assumed smaller?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Initialisation and reproducibility: the case

AI ACADEMY / AI-5/17 · 01 · LOOK AT THE FACTS

Initialisation and reproducibility: the case

The situation

Reproduce two training traces by saving their distinct initial values.

Supplied evidence

Loss $J(a)=(a-2)^2$; gradient $2(a-2)$; rate 0.25; two updates. Restart from a = 0 and separately a = 4.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain initialisation and reproducibility using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

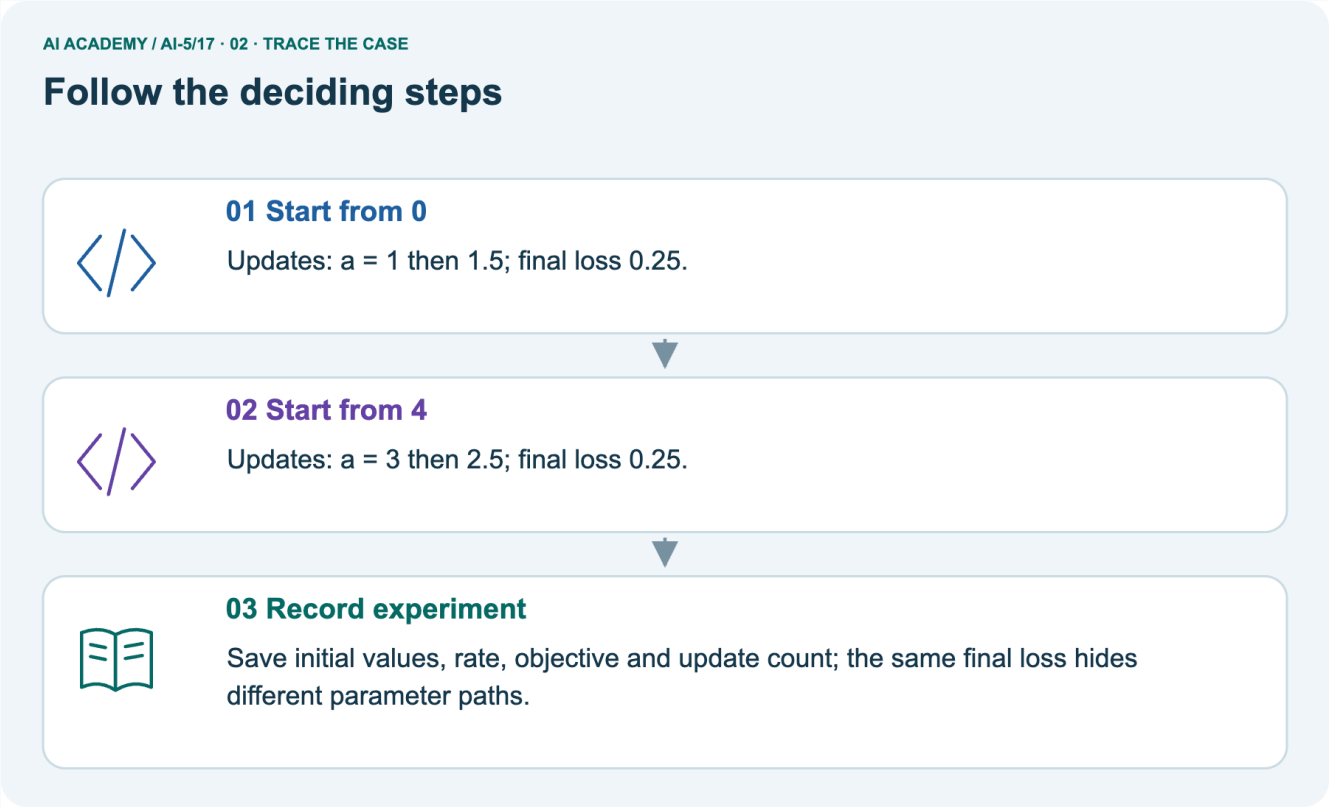
Reproduce two **training** traces by saving their distinct initial values. Supplied facts: Loss $J(a)=(a-2)^2$; gradient $2(a-2)$; rate 0.25; two updates. Restart from a = 0 and separately a = 4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Start from 0: Updates: a = 1 then 1.5; final loss 0.25. → Start from 4: Updates: a = 3 then 2.5; final loss 0.25. → Record experiment: Save initial values, rate, objective and update count; the same final loss hides different parameter paths.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Identical units receive identical updates

AI ACADEMY / AI-5/17 · 03 · INSPECT THE LESSON MODEL

Identical units receive identical updates

Hidden unit	Before the update	After the update
First ReLU unit	u is 1 and a is 1	u is 0.8 and a is 0.8
Second ReLU unit	v is 1 and b is 1	v is 0.8 and b is 0.8

In this separate two-unit example there are no biases. At x = 1, output is 2 and target is 1. Each of the four squared-error partial gradients equals 2. A simultaneous update at rate 0.1 changes every weight from 1 to 0.8 and keeps the units identical.

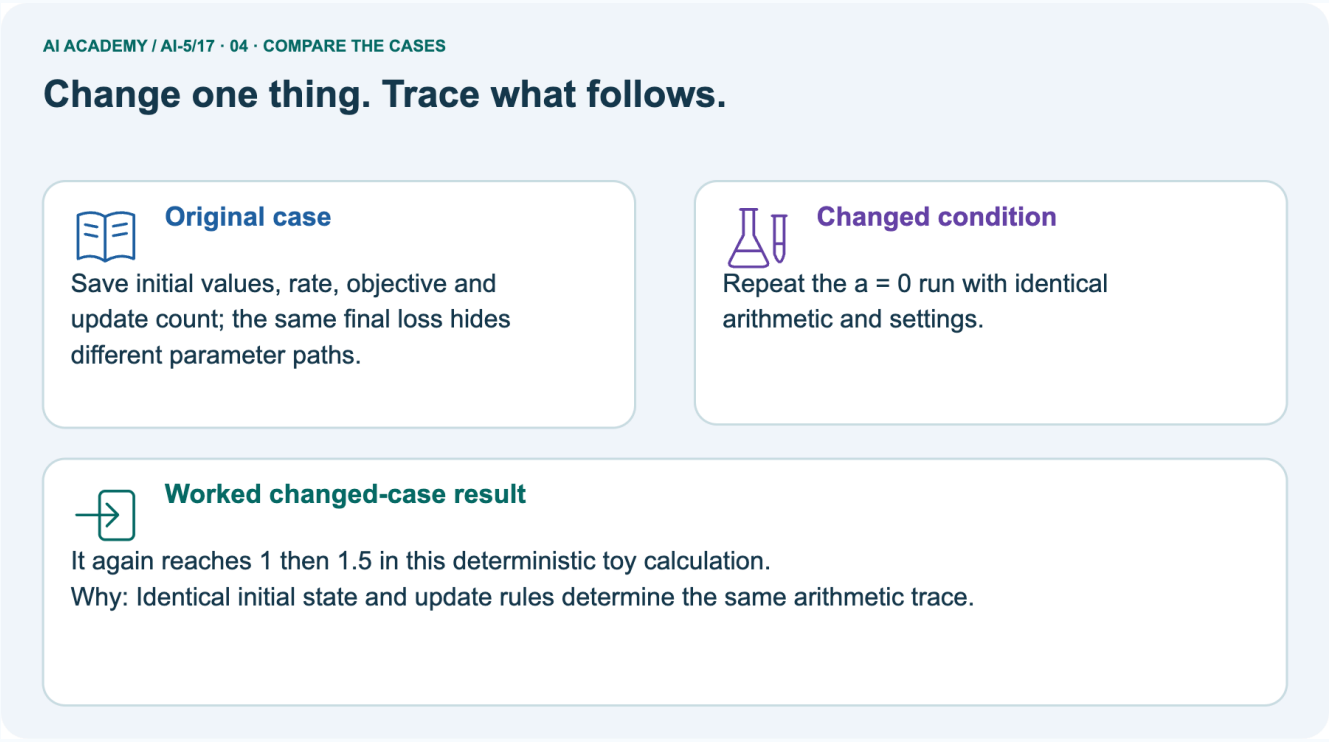
Follow the labelled picture with the matching explanation. Open the vector diagram

Identical units receive identical updates. First ReLU unit | u is 1 and a is 1 | u is 0.8 and a is 0.8. Second ReLU unit | v is 1 and b is 1 | v is 0.8 and b is 0.8

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

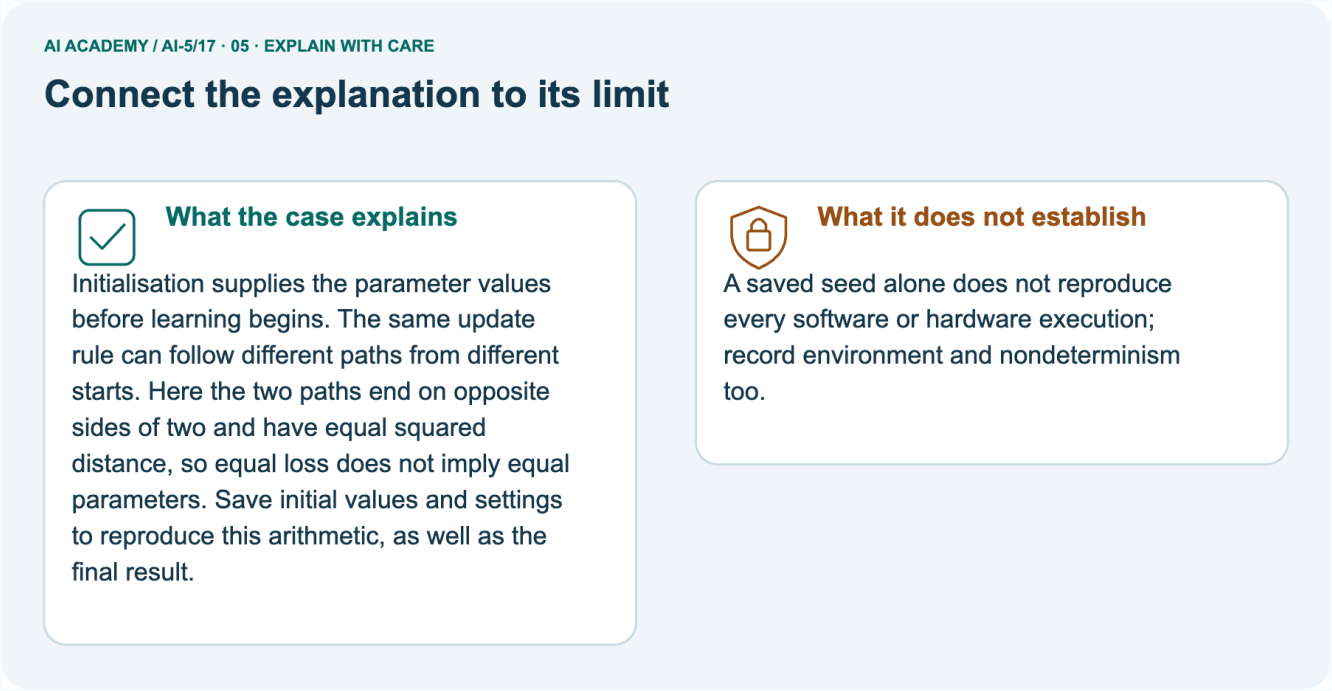
Original case: Save initial values, rate, objective and update count; the same final loss hides different parameter paths. Changed condition: Repeat the a = 0 run with identical arithmetic and settings.

Expected result: It again reaches 1 then 1.5 in this deterministic toy calculation. Reason: Identical initial state and update rules determine the same arithmetic trace.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Initialisation supplies the parameter values before learning begins. The same update rule can follow different paths from different starts.

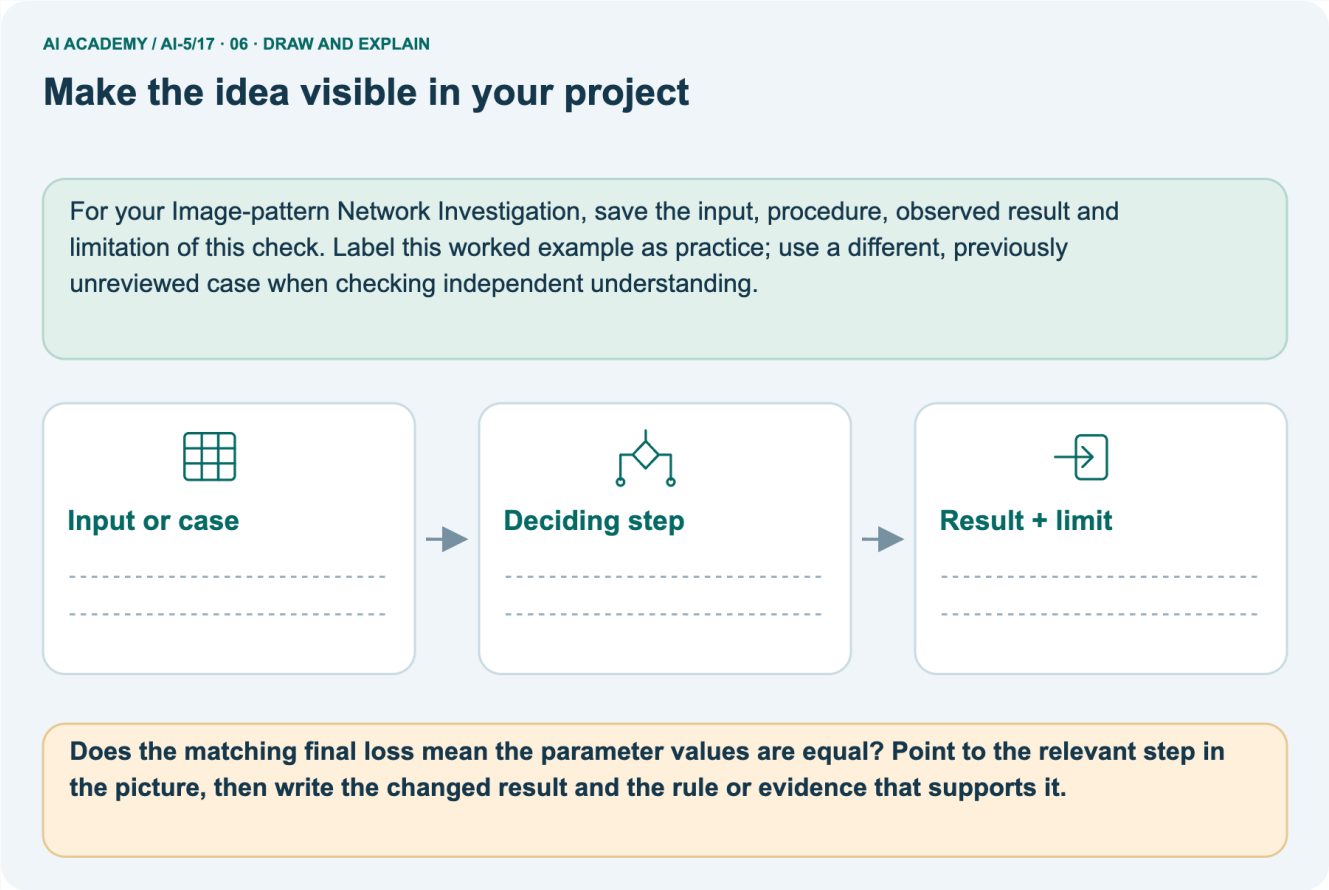
Here the two paths end on opposite sides of two and have equal squared distance, so equal loss does not imply equal parameters. Save initial values and settings to reproduce this arithmetic, as well as the final result.

Limit: A saved **seed** alone does not reproduce every software or hardware execution; record environment and nondeterminism too.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does the matching final loss mean the parameter values are equal? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Initialisation and reproducibility: the case

02. Follow the deciding steps

03. Identical units receive identical updates

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.